

Rishikesh Kumar

Full Stack Engineer | Distributed Systems

Phone: 9341421415

Email: rishi7258prince@gmail.com

LinkedIn: linkedin.com/in/rishikesh-kumar-profile

GitHub: github.com/Rishikesh-kumar-7258

Education

University	Degree	Branch	Year
IIT Bombay	M.Tech	IEOR	Jul 2024 – Jul 2026
Guru Ghasidas University	B.Tech	CSE	Jul 2020 – Jul 2024

Internships

Full-Stack Developer Intern | Vajra Labs, IIT Bombay | Powai, Maharashtra Aug 2025 – Present

- Contributed to **2 projects**, developing the **frontend of the company website** and a **sandbox-based malware analysis platform**
- Led a **3-member team**, delivering the project on schedule by coordinating tasks and implementing Agile methodologies

Full-Stack Developer Intern | Presear Software Pvt. Ltd. | Bilaspur, Chhattisgarh May 2023 – Jul 2023

- Designed a **full-stack application** using **React.js** and **Django**, following modular, component-based development practices
- Led a **2-member team** to translate client requirements into **production-ready features** and iterative product enhancements

Web Developer Intern | RMgX Technologies | Gurugram, Haryana Jun 2022 – Aug 2022

- Created and integrated **10+ fully responsive pages** for the **company website** using **Next.js**
- Collaborated with the **UI/UX team** to translate design mockups into frontend code, ensuring a cohesive and intuitive experience

Projects

Vajra Sandbox | Sandbox-based Malware Analysis and Threat Detection Platform Aug 2025 – Dec 2025

- Launched a malware analysis platform for sample submission and **static, dynamic, and memory analysis** with report generation
- Engineered a scalable system with **sandboxed multi-VM execution** and **asynchronous task pipeline** for high-throughput analysis
- Prepared software using **Python, Django/DRF, and Next.js/TypeScript** with **JWT auth, TOTP 2FA, CAPTCHA, and rate limiting**

Versatai Lab Website | Content Management Platform Dec 2025 – Present

- Created a platform to showcase research activities, student profiles, projects and events with a collaboration and feedback system
- Built the **full-stack system** using **Django/DRF** to deliver secure REST APIs and **Next.js** to power an **SSR/SSG-based frontend UI**
- Implemented **JWT-based authentication, TOTP-based 2FA, equation-based CAPTCHA protection, and role-based access control**

Sorting and Pathfinding Visualizers | Educational Algorithm Visualization Tool Dec 2022 – Jan 2023

- Built interactive **Sorting and Pathfinding Visualizers** to demonstrate algorithm behavior through **real-time algorithm execution**
- Developed using **HTML, CSS, and JS**, applying **OOP principles, data structures, and graph algorithms** to create an interface

MTP Thesis & Seminar

Parameter Specific Literature Retrieval | Master's Thesis Project (MTP) | Prof. Balamurugan Palaniappan Jul 2025 – Nov 2025

- Streamlined a **biomedical literature retrieval pipeline** to search PubMed data and extract **parameter-specific** clinical evidence
- Improved **entity extraction F1-score to 90%** by fine-tuning a **PubMedBERT-based NER model** on a manually annotated dataset
- Converted unstructured NER output into structured XML using a **BART-based Seq2Tree model** for parameter range extraction
- Implemented a **distributed ML pipeline** with **Ray, PyTorch, and HuggingFace Transformers**; deployed it with **Next.js/FastAPI**

Multimodal Transformer Architecture and Optimization | Seminar | Prof. Urban Larsson Jan 2025 – Apr 2025

- Implemented the **Attention is All You Need** paper in PyTorch and presented an in-depth analysis of the paper during the seminar
- Analyzed in-depth concepts of **An Image is Worth 16x16 Words** and presented how to use images in **transformer architecture**
- Conducted a seminar presentation on the research behind the **LoRA** paper and discussed its advantages and disadvantages

Position of Responsibility

Teaching Assistant | IE 643, Deep Learning – Theory and Practice Jul 2025 – Nov 2025

- Organized and led **weekly programming sessions** to reinforce core concepts and enhance **practical skills** in DL for **80+ students**
- Managed and **graded** student copies and projects, providing detailed **feedback** to improve their understanding of complex topics
- Facilitated **project planning** for the student teams and conducted **regular doubt sessions** ensuring project success and completion

Website Coordinator | IEOR Day, IIT Bombay Jan 2025 – Mar 2025

- Organized and coordinated** the annual **IEOR Day** with a **team of 30+ members** to enhance event visibility and resource access
- Developed and maintained the **event's official website**, providing attendees with event schedules, announcements and updates

Technical Skills

- Languages:** Python, C, C++, JavaScript, SQL, HTML, CSS, Java, Lua, TypeScript
- Frameworks & Libraries:** Django, DRF, React.js, Next.js, Node.js, Tailwind CSS, HuggingFace Transformers, PyTorch
- Systems & Tools:** PostgreSQL, Redis, Docker, Docker Compose, Git, RabbitMQ, Celery, Ray, WebSockets, Linux, REST APIs

Achievements

- Achieved **99.30 percentile (2024) and 99.13 percentile (2023)** in GATE CSE
- Secured **World Rank 32** in CodeChef Starters 42
- Secured **Rank 1** in HackForCG Hackathon by developing an **ML-based computer vision** solution using **OpenCV** and **Haar Cascade**